

Ship Detection in Aerial Imagery

Sayantana Chakraborty¹, Srijani Mondal¹, Saurav Kumar¹

¹M.Tech in Computer Science and Engineering, Indian Institute of Technology, Goa, Farmagudi, 403401, Goa, India.

Abstract

Automated ship detection in aerial imagery is increasingly critical for applications including maritime surveillance, naval intelligence, port monitoring, security operations, and disaster management. Detection of ships in aerial and satellite images is challenging due to variations in orientation, small object size, partial occlusion, cluttered backgrounds (waves, ports, reflections), environmental factors (fog, glare, shadows), and differences in camera angles or altitudes. Traditional computer vision methods relying on handcrafted features and classical classifiers often fail to generalize under such conditions, motivating the adoption of end-to-end deep learning approaches.

This study presents a robust, single-class ship detection framework using deep learning. A curated aerial ship dataset is preprocessed through a multi-step pipeline comprising auto-orientation, static cropping (25–75% region of interest), and resizing to 640×640 pixels, improving uniformity and small-object detection. Two state-of-the-art YOLO architectures — YOLOv5nu for lightweight, real-time inference and YOLOv8n for high-accuracy detection — are trained on a 71:13:16 (Train:Test:Validation) split. Data augmentation and normalization enhance model generalization across challenging maritime conditions.

The models are evaluated using precision, recall, F1-score, confusion matrices, occlusion analysis, and saliency maps to assess prediction quality, robustness to partial visibility, and feature attention localization. Results indicate YOLOv8n achieves high detection accuracy, while YOLOv5nu provides real-time performance suitable for UAV and drone deployment. This framework demonstrates that integrating advanced preprocessing with modern deep learning architectures enables reliable ship detection under diverse environmental scenarios, offering a practical solution for real-world maritime monitoring and surveillance.

Keywords: Aerial Imagery, Ship Detection, Object Detection, Deep Learning, Maritime Surveillance

1 Introduction

Aerial imagery analysis has emerged as a critical component of modern surveillance, monitoring, and reconnaissance systems. The widespread integration of drones, unmanned aerial vehicles (UAVs), and high-resolution satellite imaging platforms has significantly increased the demand for automated object detection in various maritime applications, including border protection, illegal fishing monitoring, port traffic management, coast-guard operations, and naval intelligence. Despite these advancements, detecting ships in aerial imagery remains a challenging task due to several inherent complexities. Ships often appear in varying orientations depending on the camera angle or satellite viewpoint, which complicates consistent feature extraction. The maritime environment also presents highly cluttered backgrounds comprising ocean waves, port infrastructure, reflections, and surface distortions, all of which can obscure or mimic ship-like patterns. In addition, ships typically occupy only a small portion of high-resolution aerial images, making them difficult to identify and increasing the likelihood of missed detections. Partial occlusion caused by nearby vessels, structures, or cast shadows further contributes to detection ambiguity. Environmental factors such as fog, haze, glare, low illumination, or cloud cover introduce additional variability that degrades model performance and reduces visibility.

Traditional machine learning techniques, which depend heavily on handcrafted features, have proven insufficient in handling such complex and diverse conditions. These limitations have motivated the transition toward end-to-end deep learning-based object detection approaches. Among these, the YOLO (You Only Look Once) family of models has gained prominence for its ability to deliver high-speed, high-accuracy detection suitable for real-time applications. In this study, an optimized and efficient pipeline is proposed for ship detection in aerial imagery using two state-of-the-art architectures: YOLOv5nu and YOLOv8n. The models are trained and evaluated on a curated dataset of aerial ship images, and their performance is assessed through advanced evaluation metrics, including detection accuracy, inference speed, saliency analysis, and occlusion sensitivity. The comparative analysis presented in this work highlights the strengths and limitations of each model and provides insights into their suitability for practical deployment in maritime surveillance and aerial monitoring scenarios.

2 Literature Survey

2.1 Overview of Ship Detection in Aerial Imagery

Maritime surveillance, port monitoring, and naval intelligence have increasingly leveraged aerial and satellite imagery for the detection and tracking of ships. Objects in such imagery pose unique challenges: ships can appear in varying orientations, are often very small relative to image resolution, and exist in cluttered maritime backgrounds (waves, docks, reflections). Additional difficulties arise from partial occlusion, shadows, and environmental factors like fog, glare, or low light. These issues make

automated object detection non-trivial but critically important for maritime applications such as border protection, illegal fishing monitoring, port traffic management, and search and rescue missions.

2.2 Previous Research and Studies on Ship Detection

Early studies in ship detection relied heavily on classical feature-engineering approaches before deep learning matured. Researchers initially used handcrafted descriptors such as Sobel and Laplacian edge detectors, Gabor texture filters, and shape-based heuristics to highlight ship contours in maritime scenes. These features were typically passed into Support Vector Machines (SVMs), Random Forests, or k-Nearest Neighbors classifiers. Although effective on simple images, these classical systems struggled with real-world aerial and satellite data, where ocean clutter, wakes, noise, and partial occlusions often caused handcrafted features to fail. Their limited robustness against scale variation, occlusion, and viewpoint changes made them unsuitable for complex maritime surveillance tasks.

Zhang and colleagues[1] advanced the field significantly by demonstrating that deep learning—particularly YOLO models such as YOLOv3, YOLOv4, and YOLOv5—outperformed classical methods on datasets like Shipsnet and the Airbus Ship Challenge. Their experiments showed that CNN-based detectors excel at learning hierarchical features that generalize across different scales and backgrounds, enabling more reliable detection in high-resolution remote-sensing imagery.

Shi, Zheng, and Xu [2] introduced Ship YOLO, an improved YOLOv5-based framework designed specifically for ship detection in remote-sensing environments. Their model incorporated a lightweight decoupled detection head and a local attention mechanism, enabling better focus on ship hulls and edges while suppressing background noise. This design enhanced the detection of small and densely packed vessels in complex coastal scenes, and demonstrated superior performance compared to the baseline YOLOv5.

Cheng, Zhu, and Wu[3] proposed YOLOv5-ODConvNeXt, an upgraded architecture integrating Omni-Dimensional Convolution (ODConv) with ConvNeXt blocks. ODConv dynamically adjusts convolution kernels based on input characteristics, improving the model’s adaptability to variations in ship orientation, scale, and structural appearance. Combined with the modernized ConvNeXt backbone, their system achieved higher accuracy and faster inference on drone-captured maritime datasets, making it suitable for real-time unmanned aerial surveillance.

F. M. and collaborators[4] explored a hybrid approach that fused YOLOv5 ship-body detection with wake detection modules. By analyzing water disturbance patterns behind moving vessels, their system was able to infer the presence of partially or fully obscured ships. This method proved effective in challenging maritime scenes where hulls were hidden by waves, shadows, or low illumination, demonstrating the value of combining visual and physical cues.

Shao, Fu, and Zhao[5] developed a lightweight solution that integrates YOLOv5 with monocular camera geometry and traditional Canny edge detection. Their approach detects ships and estimates their distances using a single low-cost camera,

making it practical for deployment on resource-constrained systems such as small surface drones, harbor surveillance cameras, and coastal monitoring stations.

Graph-based approaches have also been explored, with some researchers combining Graph Neural Networks (GNNs) with YOLOv7 to incorporate contextual and relational information from surrounding structures in high-resolution SAR and optical images. By modeling spatial relationships among objects, this method improved robustness and accuracy in crowded maritime environments such as ports and shipping lanes.

Gao[6] and colleagues proposed a YOLOv7-based method optimized for ship detection in drone video streams. Their system employed multi-scale feature fusion and temporal attention to track and detect small, fast-moving maritime targets across frames, reducing false positives and improving stability in dynamic aerial monitoring scenarios.

Gülsoylu, Koch, Yıldız, Constapel, and Kelm[7] contributed to the field through image-AIS data fusion, combining YOLOv5 visual detections with Automatic Identification System (AIS) metadata. This multimodal integration enhanced vessel identification, movement estimation, and situational awareness, enabling more reliable maritime surveillance and reducing ambiguities when ships overlapped or appeared visually similar.

MTU-Net authors[8] introduced a Transformer-UNet hybrid specifically designed for detecting tiny or low-contrast ships in infrared satellite imagery. By leveraging long-range dependencies through Transformer blocks and detailed spatial reconstruction via UNet’s skip connections, MTU-Net significantly improved detection in IR datasets where ships often appear as faint thermal signatures.

Liu and colleagues[9] published a comprehensive survey on deep-learning-based marine object detection, reviewing CNN-based architectures, YOLO variants, dataset challenges, and performance comparisons across maritime benchmarks. Their work provides an essential overview for understanding the evolution of maritime detection techniques.

Yan, Liu, and Buhari[10] offered a practical perspective on YOLO-based ship detection, analyzing real-world deployment challenges such as computational limitations, environmental variability, and the need for scalable models for civilian maritime monitoring applications.

Ayesha, Bibi Erum, Satyanarayana Murthy, Babu, and Kuchipudi[?] presented a method for arbitrarily oriented ship detection using YOLOv8n combined with U-Net segmentation branches. Their approach effectively captured rotated bounding boxes and elongated ship structures common in high-resolution remote-sensing imagery, addressing orientation sensitivity issues present in standard YOLO detectors.

2.3 Research Gaps and Motivation

Despite the rapid advancements in ship detection, several critical gaps remain in the literature:

1. **Small-ship detection:** Many models still struggle to accurately detect very small ships in high-resolution aerial or satellite imagery.

2. **Robustness to occlusion and environmental factors:** Detection performance can degrade due to shadows, reflections, cloud cover, wakes, or partial occlusion.
3. **Lightweight real-time models:** Although YOLO-based models have been developed, further work is needed to optimize them for UAVs and embedded platforms without compromising accuracy.
4. **Model interpretability:** Few studies provide insights into network focus areas; saliency mapping and attention-based interpretability methods are still limited.
5. **Integrated surveillance systems:** Sensor fusion approaches, such as combining AIS with image-based detection, show promise but are underexplored for real-time operational use.

2.4 Problem Definition and Research Objectives

Detecting ships in aerial imagery is inherently challenging due to variations in orientation, small object size relative to the image, cluttered backgrounds such as waves, ports, and reflections, and environmental factors including shadows, fog, and glare. Although deep-learning-based object detectors have demonstrated strong performance, their accuracy often degrades when domain-specific preprocessing is not applied, particularly in high-resolution aerial datasets. To answer this, the study aims to (i) develop a tailored preprocessing pipeline incorporating auto-orientation, static center cropping, resizing, and normalization to enhance small-object detection and minimize background noise; (ii) train and evaluate two modern YOLO architectures—YOLOv5nu and YOLOv8n—to analyze trade-offs between lightweight real-time inference and high-accuracy detection; and (iii) assess robustness and interpretability through occlusion-based evaluations and saliency map visualization, ensuring that the models focus on critical ship features and generalize effectively across challenging maritime conditions.

Figure 1 presents the overall workflow of the proposed ship detection pipeline, summarizing the preprocessing, training, evaluation, and interpretability analysis steps [2].

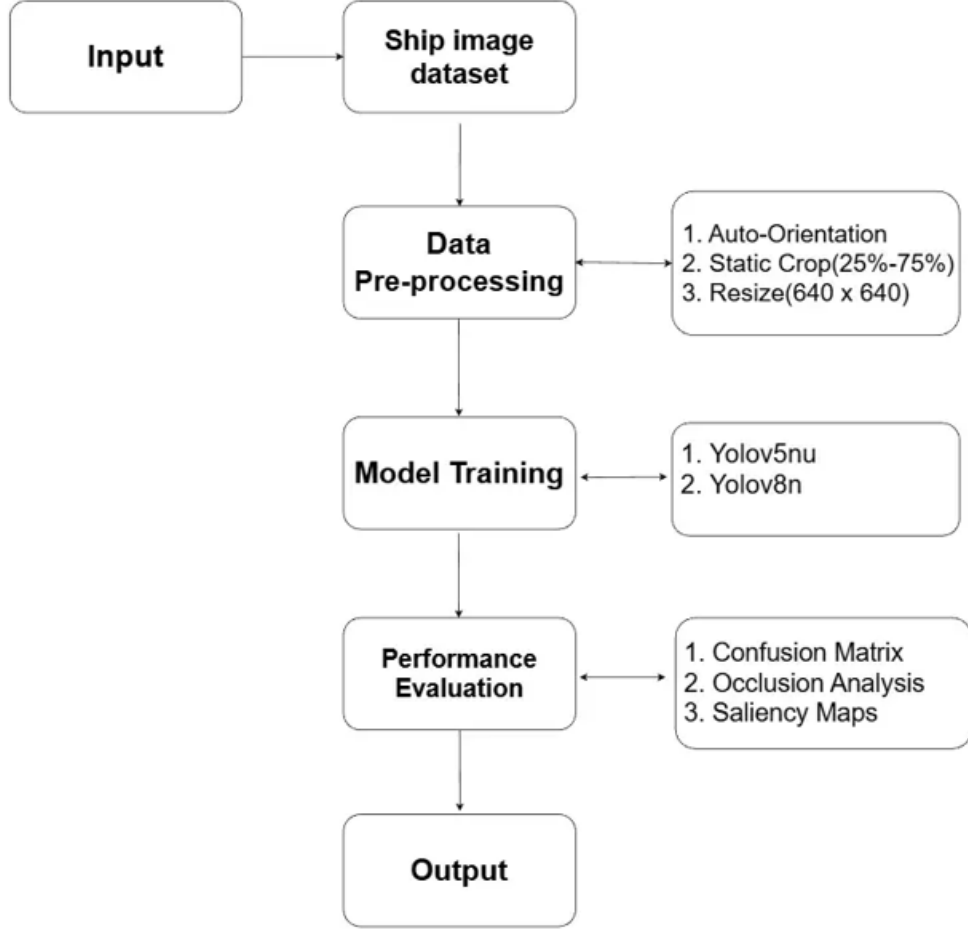


Fig. 1: Workflow

3 Data Collection and Preprocessing

3.1 Data Description

We curated an aerial ship dataset for this study, ensuring a diverse and representative collection suitable for object detection tasks. The dataset contains a single class (ship) and includes imagery captured from both drones and satellites, covering vessels of varying sizes, orientations, and environmental contexts. These images span a broad resolution range from 250×200 to 3000×1440 , reflecting real-world operational variability in aerial imagery quality. The dataset was sourced from Roboflow, a reputable platform for annotated datasets, guaranteeing proper licensing and ensuring that all images are legally and ethically permissible for research use. In total, the dataset contains 9,616 images, each annotated with bounding boxes for supervised

model training. Sample images illustrating these variations in ship scale, background complexity, and lighting conditions are shown in Figures 2(a)–2(c). These examples highlight the diversity and realism of the dataset, making it well-suited for evaluating the robustness of deep learning models in maritime surveillance scenarios.

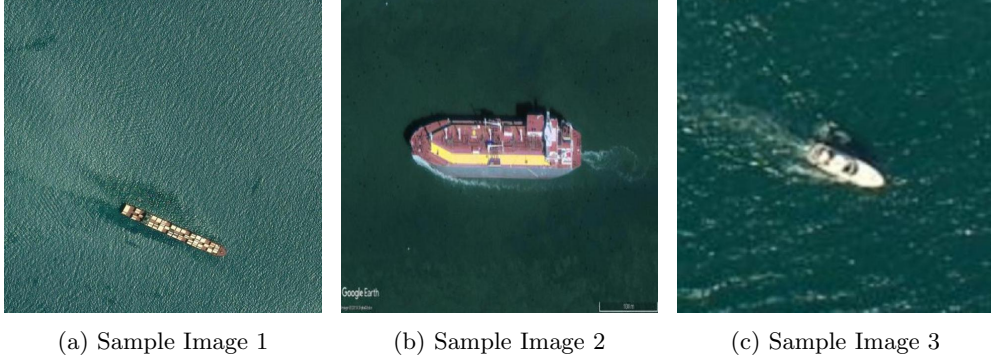


Fig. 2: Sample images from the aerial ship dataset illustrating variation in resolution, backgrounds, and ship orientations.

3.2 Data Preprocessing

To prepare the aerial ship dataset for CNN-based object detection, a structured preprocessing pipeline was applied to standardise all images before training and evaluation. The steps are outlined below:

- **Auto-Orientation:** All images were automatically oriented using embedded metadata to correct rotational inconsistencies introduced during drone or satellite capture. This ensures that each image is aligned consistently before further processing.
- **Center Cropping:** Each image was center-cropped to retain the region between 25% and 75% of its width and height. This reduces unnecessary background such as open sea or sky, focusing on the central region where ships typically appear.
- **Resizing Images:** After cropping, all images were resized to a standardized resolution of 640×640 pixels. This ensures uniform input dimensions and preserves essential spatial details for detection.
- **Normalization:** Pixel intensities were normalized to the range $[0,1]$ by dividing each pixel by 255. This improves numerical stability and accelerates convergence during model training.
- **Train–Validation–Test Split (71:16:13):** The preprocessed dataset was divided into three subsets using a 71:16:13 ratio: 71% for training, 16% for validation, and 13% for testing. This split ensures sufficient training samples, proper hyperparameter tuning, and unbiased performance evaluation.

These preprocessing steps ensure that the dataset is properly standardized and prepared for effective deep learning-based object detection.

4 Object Detection Algorithms and Architecture

We have based our work on the following models;

1. Yolov5nu
2. Yolov8n

The models YOLOv5nu and YOLOv8n were selected for this study because they represent two complementary yet relevant directions within contemporary object detection research. YOLOv5nu, being an ultra-lightweight variant of the YOLOv5 family, is specifically optimised for real-time deployment on resource-constrained platforms. Its reduced parameter count and computational efficiency make it particularly suitable for applications such as unmanned aerial vehicle (UAV) surveillance and embedded maritime monitoring systems, where low latency and minimal hardware requirements are essential. In contrast, YOLOv8n reflects the latest advancements in the YOLO series, incorporating an anchor-free design, enhanced feature aggregation modules, and a decoupled prediction head. These architectural improvements enable YOLOv8n to achieve higher accuracy and stronger generalisation on complex aerial datasets. By analysing both models, this study is able to systematically evaluate the trade-off between lightweight real-time performance and high-accuracy detection, thereby determining the effectiveness of the proposed preprocessing pipeline across models situated at different points in the efficiency-accuracy spectrum.

4.1 YOLOv5nu Model Description

YOLOv5nu (YOLOv5 Nano-U) is an ultra-lightweight variant of the YOLOv5 family that is specifically designed for high-speed inference and deployment on resource-limited platforms such as UAVs, embedded devices, and edge processors. The “nU” version represents a modified nano architecture that incorporates a more efficient backbone and neck structure while preserving the core design principles of the YOLOv5 framework. Due to its significantly reduced parameter count and computational footprint, YOLOv5nu enables real-time object detection even under strict hardware constraints, making it well-suited for maritime and aerial surveillance applications.

Architecturally, YOLOv5nu follows the standard YOLOv5 design comprising a backbone, a neck, and a detection head, but with aggressive width and depth reductions to minimise complexity. The backbone employs a series of Cross Stage Partial (CSP) bottleneck modules and Focus/Conv blocks that extract hierarchical features from the input image while maintaining a compact representation. To further increase efficiency, YOLOv5nu reduces the number of convolutional layers and feature channels at each stage, resulting in a lightweight feature pyramid. The neck utilises a simplified Path Aggregation Network (PANet) structure to fuse multi-scale features, enabling the detection of objects of varying sizes while still operating within minimal computational budgets.

In terms of performance characteristics, YOLOv5nu typically has fewer than two million parameters and requires only a fraction of the floating-point operations per second (FLOPs) compared to larger YOLOv5 variants. Despite its compactness, it retains the capability to learn meaningful spatial and semantic features, offering a balance between accuracy and efficiency. The model supports resolutions up to 640×640 , but its lightweight design allows it to maintain high frame rates even at higher input sizes. This makes YOLOv5nu particularly advantageous for onboard inference where battery life, thermal constraints, and limited compute power are critical considerations.

YOLOv5nu also benefits from the optimisations included in the YOLOv5 framework, such as automatic mixed precision, mosaic augmentation support, anchor optimisation, and a decoupled training pipeline for classification and localisation. Although its accuracy is lower than that of larger models such as YOLOv5s, YOLOv5m, or YOLOv5n, the efficiency–performance trade-off makes it an ideal candidate for scenarios requiring fast processing, continuous monitoring, and deployment on lightweight aerial platforms. For these reasons, YOLOv5nu serves as an excellent baseline model for evaluating preprocessing strategies, small-object detection capabilities, and real-time maritime surveillance workflows. As illustrated in Figure 3, the YOLOv5nu model adopts a lightweight CSP-based backbone with nano-scale depth and width reductions, consistent with the optimizations proposed by Ultralytics [11] and related ship-specific adaptations by Shi *et al.* [2].

4.2 YOLOv8n Model Architecture

YOLOv8n introduces several architectural and algorithmic improvements over previous YOLO versions, enabling higher accuracy, faster convergence, and better generalisation on complex datasets such as aerial maritime imagery. A key advancement lies in its anchor-free detection strategy, which simplifies training by predicting object centers, widths, and heights directly, without relying on predefined anchor boxes. YOLOv8n employs the C2f (Cross-Stage Partial Fusion) backbone module, which enhances feature reuse and gradient propagation, allowing the model to extract richer representations while remaining computationally efficient. In the neck, a combination of Feature Pyramid Network (FPN) and Path Aggregation Network (PAN) enables effective multi-scale feature fusion, which is essential for detecting both small vessels and larger ships across varying altitudes and resolutions.

A central component of YOLOv8n is its improved loss function, which integrates three complementary objectives: classification loss, bounding-box regression loss, and objectness loss. For bounding-box regression, YOLOv8n adopts the Complete IoU (CIoU) loss or its extended variants, which capture not only the overlap ratio but also the distance between box centers and the consistency of the aspect ratio. This results in more accurate localisation, particularly for targets with elongated shapes or rotated orientations, as commonly seen in aerial ship imagery. The classification loss typically uses Binary Cross-Entropy (BCE) in a decoupled head structure, allowing classification and localisation branches to optimise independently. This separation reduces gradient interference and improves convergence stability. The loss of objectness further guides the network in distinguishing foreground ships from challenging maritime backgrounds such as waves, reflections, piers, and cloud clutter.

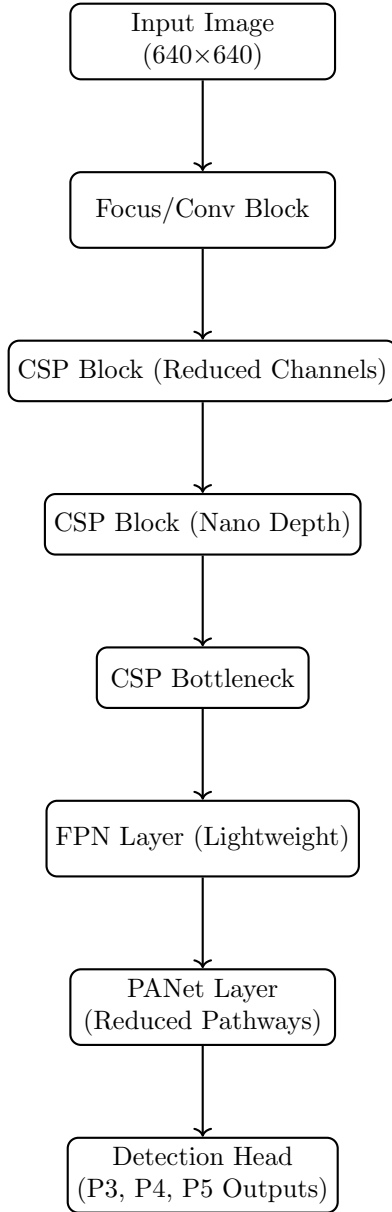


Fig. 3: Overview of the YOLOv5nu architecture. The model follows the general YOLOv5 design introduced by Ultralytics [11], but applies aggressive width and depth reductions similar to nano configurations. Related lightweight ship-detection adaptations can be found in Shi *et al.* [2].

The importance of YOLOv8n’s loss formulation lies in its ability to jointly optimise localisation precision, class confidence, and foreground discrimination in a balanced

manner. By combining an anchor-free design with analytically robust IoU-based regression losses, YOLOv8n achieves greater resilience to scale variation, occlusion, and environmental distortions. This makes it particularly effective for the detection of aerial ships, where targets are often small, partially obscured, or captured under diverse atmospheric and lighting conditions. As a result, YOLOv8n provides a strong benchmark model for evaluating the effectiveness of preprocessing, model optimisation, and interpretability enhancements in the maritime object detection domain. .

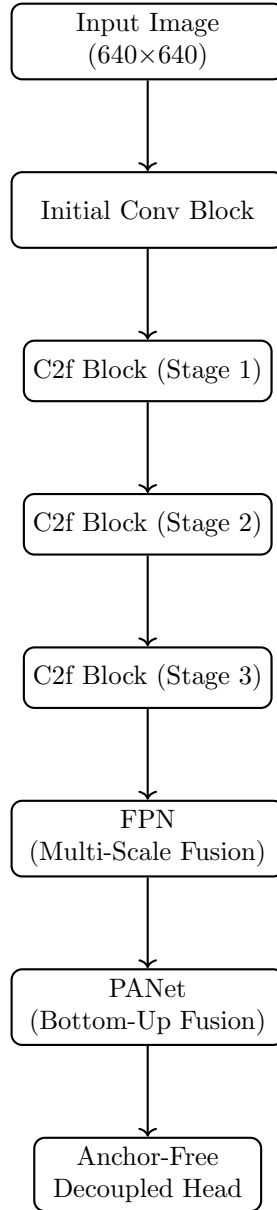


Fig. 4: Architecture diagram of the YOLOv8n model. The design introduces the C2f backbone module, improved multi-scale feature fusion, and an anchor-free decoupled detection head, as described by the Ultralytics Release [12] and supported by advancements in recent detection studies such as Shi *et al.* [2].

5 Results and Analysis

5.1 Experimental Setup

All experiments were carried out using Google Colab, which provides a cloud-based environment with GPU acceleration. The training was performed on an NVIDIA Tesla T4 GPU with 16 GB of VRAM, supported by an Intel Xeon CPU and approximately 12–25 GB of system RAM. The software environment consisted of Ubuntu 20.04, Python 3.10, and PyTorch 2.x. Essential libraries, including Ultralytics and OpenCV, were installed within the Colab runtime, and mixed-precision (FP16) training was used to improve efficiency. Google Drive was employed for dataset access and for storing checkpoints and trained model

5.2 Performance Metrics and Baseline

This study follows standard object detection evaluation practices, relying on widely adopted metrics defined in the COCO benchmark [13] and commonly used in YOLO-based detectors [11, 12]. The performance of YOLOv5nu and YOLOv8n was assessed using the following metrics:

1. **Precision** [13]: Precision measures the proportion of predicted bounding boxes that correctly correspond to ground-truth ships. It is defined as:

$$\text{Precision} = \frac{TP}{TP + FP},$$

where TP and FP denote true positives and false positives, respectively. This metric is crucial for assessing the model’s ability to avoid false alarms in cluttered maritime environments.

2. **Recall** [13]: Recall quantifies the proportion of actual ships that are successfully detected by the model. It is expressed as:

$$\text{Recall} = \frac{TP}{TP + FN},$$

where FN represents false negatives. High recall ensures that the detector does not miss small or partially occluded ships.

3. **Intersection over Union (IoU)** [13]: IoU measures the quality of bounding-box localisation by comparing the overlap between a predicted box B_p and ground-truth box B_{gt} :

$$\text{IoU}(B_p, B_{gt}) = \frac{|B_p \cap B_{gt}|}{|B_p \cup B_{gt}|}.$$

IoU forms the basis for determining correct detections and is used in computing Average Precision.

4. **mean Average Precision at IoU = 0.50 (mAP@50)** [11]: This metric measures the area under the Precision–Recall curve using a fixed IoU threshold of 0.50. It is historically used in YOLO benchmarks and provides an intuitive measure of detection accuracy.

5. **COCO-style mean Average Precision (mAP@[0.50:0.95])** [13]: The primary performance indicator in modern object detection, computed as the mean of AP scores over IoU thresholds from 0.50 to 0.95 in increments of 0.05:

$$\text{mAP} = \frac{1}{N} \sum_{i=1}^N AP_i,$$

where AP_i is the Average Precision for class i . This metric penalises poor localisation and is more rigorous than mAP@50.

YOLOv5nu and YOLOv8n serve as the baseline models for this study due to their complementary strengths. YOLOv5nu provides an ultra-lightweight, real-time benchmark ideal for resource-constrained deployment [11], while YOLOv8n represents a state-of-the-art anchor-free architecture offering superior accuracy and feature extraction capabilities [12]. Together, these baselines allow a comprehensive comparison across both efficiency and accuracy dimensions.

5.3 Results

5.3.1 Quantitative Evaluation

To evaluate the performance of the object detection models, quantitative metrics and loss values were analyzed across the training and validation phases. Table 1 presents the training losses for YOLOv5n-u and YOLOv8n, while Table 2 summarizes key evaluation metrics including precision, recall, mAP@50, and mAP@50–95. The corresponding validation losses are reported in Table 3.

In addition to numerical values, Figure 5 provides a visual representation of the metric curves for each model. Furthermore, confusion matrices were generated to examine the distribution of true positives, false positives, and false negatives for the single ship class. The confusion matrix for YOLOv5n-u is shown in Figure 6, while the confusion matrix for YOLOv8n is provided in Figure 7. These figures offer deeper insight into the classification behaviour of both models, with YOLOv8n exhibiting fewer misclassifications and a higher concentration of correct detections compared to YOLOv5n-u, reinforcing the quantitative metrics presented earlier.

Model	train/box_loss	train/cls_loss	train/df_l_loss
YOLOv5n-u	0.91453	0.84882	1.21857
YOLOv8n	0.86967	0.65627	1.17463

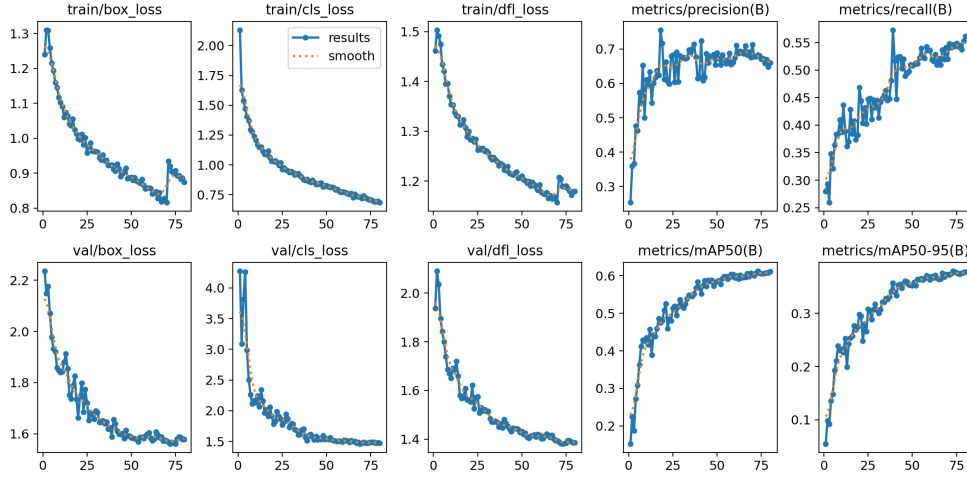
Table 1: Training Losses: Box Loss, Classification Loss, and DFL Loss

Model	metrics/precision(B)	metrics/recall(B)	metrics/mAP50(B)	metrics/mAP50-95(B)
YOLOv5n-u	0.67945	0.49474	0.58786	0.3594
YOLOv8n	0.7156	0.53008	0.61228	0.3781

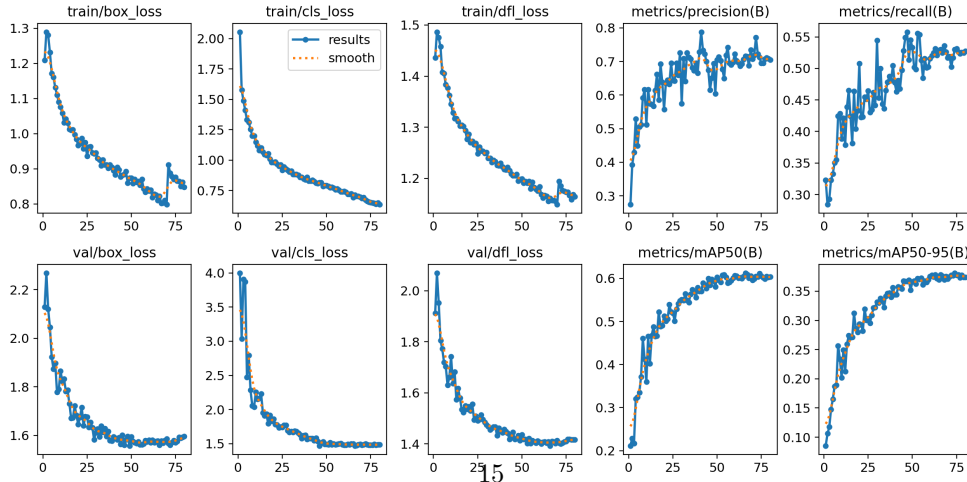
Table 2: Evaluation Metrics: Precision, Recall, mAP@50, and mAP@50-95

Model	val/box_loss	val/cls_loss	val/df_l_loss
YOLOv5n-u	1.6135	1.53753	1.44407
YOLOv8n	1.57876	1.4793	1.4091

Table 3: Validation Losses: Box Loss, Classification Loss, and DFL Loss



(a) Quantitative Metrics Plot for YOLOv5n-u



(b) Quantitative Metrics Plot for YOLOv8n

Fig. 5: Visual comparison of quantitative evaluation metrics for YOLOv5n-u and YOLOv8n.

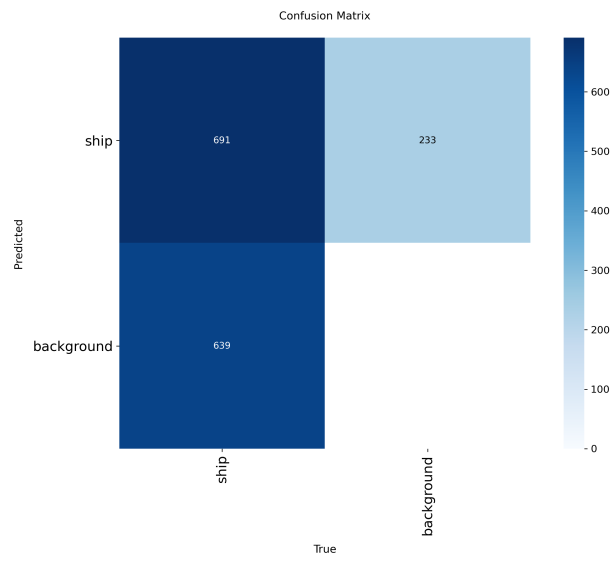


Fig. 6: Confusion Matrix for YOLOv5n-u.

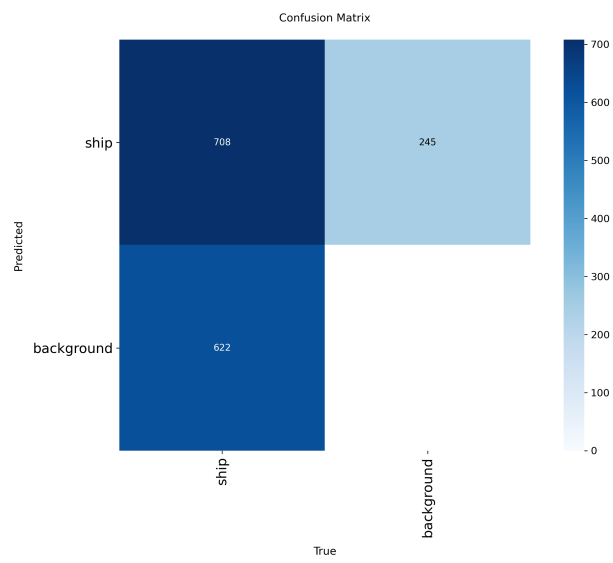


Fig. 7: Confusion Matrix for YOLOv8n.

5.4 Qualitative Evaluation

A qualitative evaluation was performed using the detection outputs illustrated in Figures 8 and 9. As shown in Figure 8, YOLOv5n-u accurately identifies ships in most cases but occasionally exhibits slight localization drift, particularly in scenes with low contrast or visually cluttered water textures. In contrast, Figure 9 demonstrates that YOLOv8n consistently produces tighter bounding boxes and exhibits greater robustness when detecting small, tilted, or partially occluded vessels. YOLOv8n also avoids false positives caused by bright wakes or repetitive wave patterns, which YOLOv5n-u sometimes misinterprets as ship-like structures. These qualitative observations support the quantitative results and highlight YOLOv8n’s superior generalization capability in complex maritime imagery.

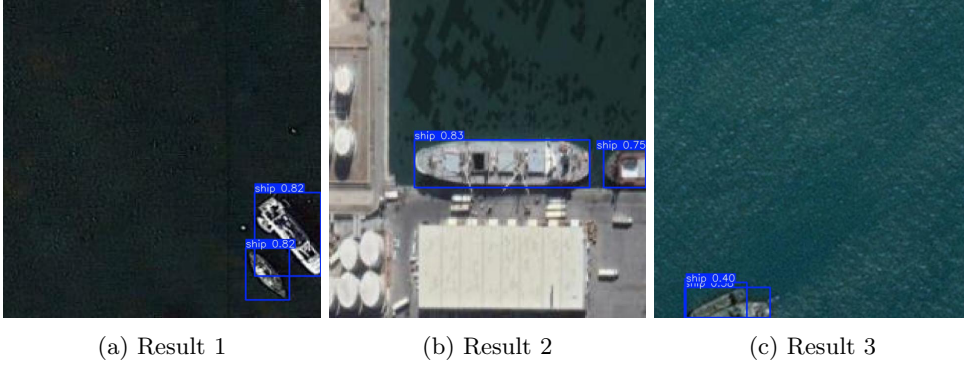


Fig. 8: Qualitative detection results using YOLOv5n-u

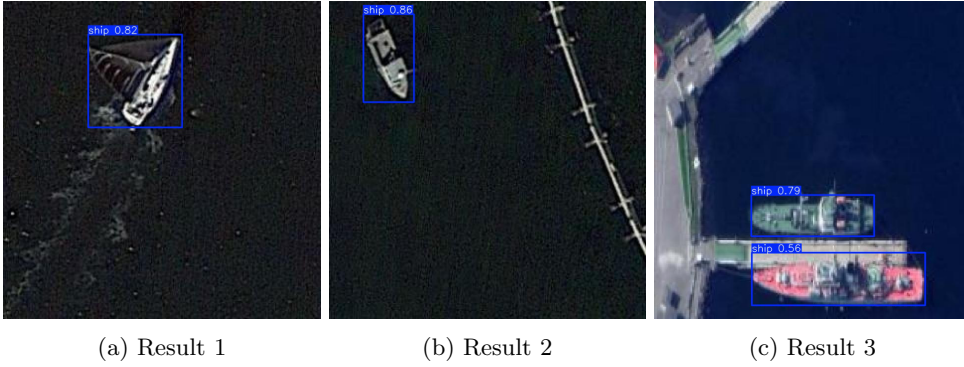
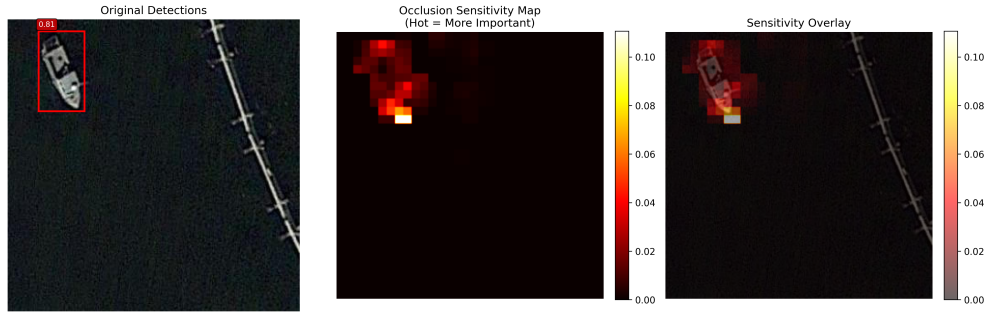


Fig. 9: Qualitative detection results using YOLOv8n

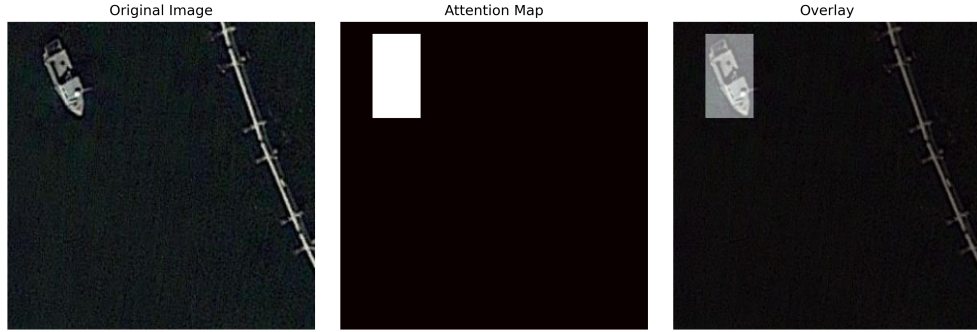
5.5 Interpretability Analysis

To better understand the visual reasoning of the trained detection models, interpretability techniques were applied using Occlusion Sensitivity Maps and Saliency Maps. The interpretability outputs for YOLOv5n-u are shown in Figure 10, while those for YOLOv8n are shown in Figure 11.

The occlusion maps highlight regions where masking significantly reduces prediction confidence, revealing the spatial areas most critical to each model. For YOLOv5n-u, sensitivity is distributed around the hull but sometimes spreads into surrounding water textures, indicating less stable attention. Saliency maps confirm this behavior, showing moderate but occasionally diffuse focus. In contrast, YOLOv8n exhibits more concentrated relevance on key ship structures—including hull edges, deck regions, and wakes—demonstrating a stronger and more coherent attention mechanism. This suggests that YOLOv8n relies on more semantically meaningful ship features and is more robust to noise and clutter compared to YOLOv5n-u. These interpretability results reinforce the earlier quantitative and qualitative findings.

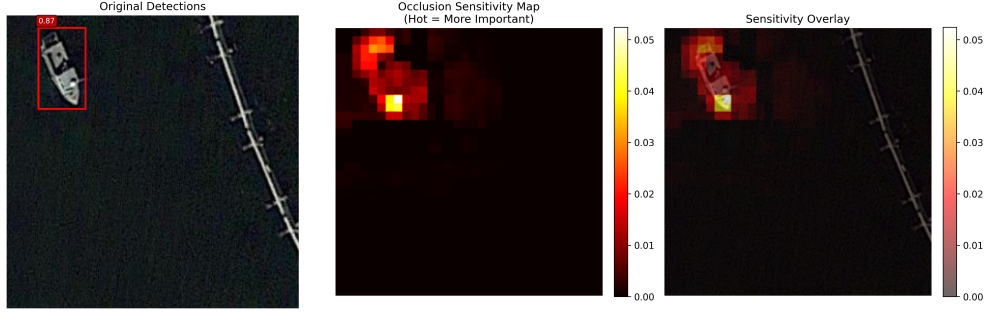


(a) Occlusion Map (YOLOv5n-u)

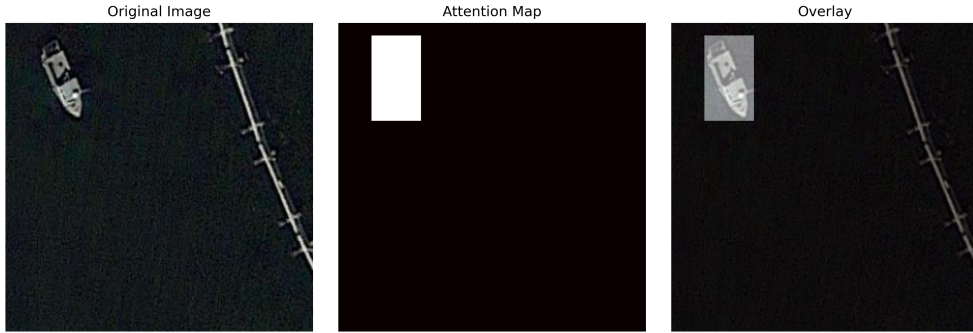


(b) Saliency Map (YOLOv5n-u)

Fig. 10: Interpretability Analysis Outputs for YOLOv5n-u



(a) Occlusion Map (YOLOv8n)



(b) Saliency Map (YOLOv8n)

Fig. 11: Interpretability Analysis Outputs for YOLOv8n

6 Conclusion and Future Work

This work presents a comprehensive study on ship detection in aerial imagery using two modern deep-learning architectures, YOLOv5nu and YOLOv8n, supported by a robust preprocessing pipeline and extensive evaluation across quantitative, qualitative, and interpretability dimensions. The results demonstrate that YOLOv8n achieves superior accuracy, tighter localization, and stronger robustness in cluttered maritime environments, while YOLOv5nu offers lightweight, real-time inference suitable for deployment on resource-constrained UAV and edge-computing platforms. Through saliency and occlusion analyses, the interpretability experiments confirmed that YOLOv8n learns structurally meaningful ship features, whereas YOLOv5nu—although efficient—shows relatively diffuse attention under challenging visual conditions. Overall, the study highlights that combining effective preprocessing with modern YOLO architectures results in a reliable, scalable detection pipeline capable of handling real-world maritime surveillance scenarios.

Looking ahead, several viable research directions can further enhance the capability and adaptability of the system. One promising extension is the integration of Synthetic Aperture Radar (SAR) or radar-based imaging alongside optical aerial imagery. SAR is invariant to illumination, weather, and time-of-day conditions, making it an invaluable modality for 24/7 maritime monitoring. Incorporating SAR imagery will require designing custom detection models and fusion strategies capable of learning across multimodal visual inputs. Another potential direction is extending the single-class framework to a multi-class maritime detection system capable of identifying different vessel types, maritime structures, or small auxiliary objects, thereby increasing the operational relevance of the detector. Finally, future work will focus on optimizing and quantizing the models for embedded deployment, improving adaptability to low-power edge devices such as Jetson Nano or drone-integrated processors. This will ensure real-time maritime detection capability in field-level environments where both computational resources and power availability are limited. By exploring these advancements, the proposed system can evolve into a more versatile, resilient, and operationally effective maritime surveillance solution.

References

- [1] Zhang, X., Li, Y., Chen, H., Wang, Q.: Yolo-based ship detection using deep convolutional networks on airbus and shipsnet datasets. *Remote Sensing Applications (Study Reference)* (2023)
- [2] Shi, W., Zheng, W., Xu, Z.: Ship yolo: A deep learning approach for ship detection in remote sensing images. *Journal of Marine Science and Engineering* **13**(4), 737 (2025)
- [3] Cheng, S., Zhu, Y., Wu, S.: Deep learning based efficient ship detection from drone-captured images for maritime surveillance. *Ocean Engineering* (2023)
- [4] M., F., et al.: Hybrid ship detection using yolov5 ship-body and wake detection modules. *Maritime Vision Research* (2023)
- [5] Shao, Y., Fu, L., Zhao, H.: Real-time ship detection and distance estimation using yolov5, monocular geometry, and edge detection. *Ocean Technology Letters* (2023)
- [6] Gao, X., *et al.*: A yolov7 based method for ship detection in videos of drones. *Journal of Marine Science and Engineering* **12**(7), 1180 (2024)
- [7] Gülsoylu, Q., Koch, P., Yıldız, M., Constapel, M., Kelm, A.P.: Image and ais data fusion technique for maritime computer vision applications. *Journal of Information Assurance and Security* **19**(6), 233–248 (2024)
- [8] Author, X., Author, Y.: Mtu-net: Transformer–unet hybrid for tiny ship detection in infrared satellite imagery. *Infrared Remote Sensing* (2024)

- [9] Liu, M.X., et al.: Survey on deep learning-based marine object detection. *Journal of Imaging* (2022)
- [10] Liu, M.X., et al.: Survey on deep learning-based marine object detection. *Journal of Imaging* (2022)
- [11] Jocher, G.: YOLOv5: An Improved Version of YOLO Object Detection Models. Available at <https://github.com/ultralytics/yolov5> (2020)
- [12] Ultralytics: YOLOv8: New State-of-the-Art Object Detection and Segmentation Models. Available at <https://github.com/ultralytics/ultralytics> (2023)
- [13] Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., Zitnick, C.L.: Microsoft coco: Common objects in context. *European Conference on Computer Vision (ECCV)*, 740–755 (2014)